LOTQ-0006 / CORE MATH / BEGINNER / FIELD GUIDE

Sequences, Series, and Financial Growth

Ordered values, summed values, compounding, discounting, and equity-curve mechanics.

Library of the Quant

Educational reference manual

Manual Profile

LIBRARY ID	LOTQ-0006
CATEGORY	Core Math
DIFFICULTY	Beginner
FORMAT	Single-column field-guide manual
USE	Study, lookup, and research-system reference

Educational Use Only

How to Use This Manual

READING PATH

Read each page top to bottom. The design is intentionally single-column: meaning first, mechanics second, quant use third, implementation and pitfalls last. No parallel article columns are used.

STUDY LOOP

For each card: say the one-sentence meaning out loud, trace the formula with numbers, connect it to a market example, then ask how it could break inside a backtest.

WHAT TO FOCUS ON

Manual element	Use it for	Question to ask
Formula blocks	Remember the exact growth, sum, or discount equation	What does each symbol represent?
Worked calculations	See the arithmetic without hiding behind jargon	Could I reproduce this in Python?
System-fit boxes	Connect math to data pipelines, features, backtests, and reports	Where would this object live in my research stack?
Failure-mode notes	Avoid beginner mistakes that create fake performance	What false conclusion could this cause?

NOTATION CONVENTION

This guide uses t for time index, n for a generic integer step, P_t for price at time t, r_t for simple return, $g_t = 1 + r_t$ for growth factor, and W_t for wealth or equity.

FAST REVIEW METHOD

When revisiting the PDF, skip to the formula summary and review pages first, then return to individual concept cards that feel weak. This keeps review active instead of passive.

Notation and Formula Map

CORE NOTATION TABLE

Symbol	Plain-English meaning	Quant interpretation
n	Step count or term number	Generic index in a sequence or series.
t	Time index	Trading day, bar, month, event number, or rebalance step.
a_n	nth term of a sequence	The value observed or generated at step n.
S_n	Partial sum through n terms	Running total, cumulative cash flow, or accumulated series.
P_t	Price at time t	Input price used to compute returns or features.
r_t	Simple return	Percent change from P_{t-1} to P_t.
g_t	Growth factor: $1 + r_t$	Multiplier applied to wealth.
W_t	Wealth or equity at time t	Backtest account value or portfolio value.

EQUATION CALLOUT

```
Sequence: a_1, a_2, ..., a_n Series: S_n = a_1 + a_2 + ... + a_n | Sequence lists values; series adds them.
```

EQUATION CALLOUT

```
Compounding: W_n = W_0 x product_{t=1..n}(1+r_t) | Growth is multiplicative, not additive.
```

QUANT LENS

Most research objects are sequences first: prices, returns, signals, positions, trade PnL, volatility estimates, and equity curves. Series appear when you aggregate those ordered values into totals, averages, discounted cash flows, or cumulative performance.

BEGINNER WARNING

A sequence keeps order. A plain sum can erase order. In finance, order often matters because compounding, drawdowns, leverage, margin, and investor withdrawals are path-dependent.

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Manual map

Each entry is a full-width, single reading path reference card. Page numbers include title and front matter.

Table of Contents

Manual map

Each entry is a full-width, single reading path reference card. Page numbers include title and front matter.

CORE MATH / BEGINNER / SEQUENCE BASICS

Sequences as Ordered Values

ONE-SENTENCE MEANING

A sequence is a list of values arranged in a defined order, usually indexed by n or by time t.

MEMORY JOGGER

Think: not just a bag of numbers - a line of numbers with a first, next, and later.

FORMULA / KEY MECHANICS

```
a = (a_1, a_2, ..., a_n) time-indexed form: x_t for t = 1, 2, ..., T Write a sequence as a_1,
a_2, a_3, ... or as a_t when the index is time. The index is part of the information: a
return on Monday and the same return on Friday can have different meaning because surrounding
values changed.
```

REFERENCE TABLE

Object	Sequence example	Why order matters
Price	P_1, P_2, ..., P_T	Returns depend on adjacent prices.
Signal	s_1, s_2, ..., s_T	Trading action depends on when signal appeared.
Equity	W_1, W_2, ..., W_T	Drawdown depends on peaks and later losses.

WHY IT MATTERS IN QUANT RESEARCH

Prices, returns, signals, orders, fills, spreads, volumes, and model predictions all arrive as sequences before they become statistics.

SIMPLE MARKET / TRADING EXAMPLE

A daily close series [100, 102, 101, 105] is a sequence. Its order tells you there was a dip before the move to 105.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Store each sequence with timestamp, asset id, frequency, source, and adjustment rules. Without those fields, later features and backtests become ambiguous.

CONFUSIONS TO AVOID

Do not treat a sequence like an unordered sample when path matters. Sorting returns by size destroys the original market path.

RELATED TERMS / QUICK RECALL

Indexing, time series, return stream, path dependency, data pipeline.

CORE MATH / BEGINNER / SEQUENCE BASICS

Indexing: n, t, and Market Time

ONE-SENTENCE MEANING

Indexing assigns a position to each value so the sequence can be referenced, shifted, aligned, and compared.

MEMORY JOGGER

Think: the index is the address of a value.

FORMULA / KEY MECHANICS

```
Lag: x_{t-1} Lead: x_{t+1} Return: r_t = P_t / P_{t-1} - 1 Use n for abstract math steps and t for time. In market data, t may mean calendar day, trading day, bar number, trade event, or rebalance period. These are not interchangeable.
```

REFERENCE TABLE

Index choice	Means	Common bug
n	Generic term number	Forgetting that n has no calendar meaning.
t	Time step	Using t before defining bar frequency.
event index	Trade or quote number	Treating irregular events like equal time bars.

WHY IT MATTERS IN QUANT RESEARCH

Bad indexing creates look-ahead bias, misaligned features, wrong returns, and invalid joins between prices, fundamentals, and signals.

SIMPLE MARKET / TRADING EXAMPLE

If P_t is today close, a signal computed from P_t should usually trade at t+1 or later, not at the same close unless the execution assumption is explicit.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Every table should define its index granularity: timestamp timezone, session calendar, asset universe, and whether bars are end-stamped or start-stamped.

CONFUSIONS TO AVOID

Confusing calendar days with trading days silently breaks rolling windows around holidays and missing bars.

RELATED TERMS / QUICK RECALL

Lag, lead, shift, alignment, timestamp, trading calendar.

CORE MATH / BEGINNER / SEQUENCE BASICS

Arithmetic Sequences

ONE-SENTENCE MEANING

An arithmetic sequence changes by adding the same fixed amount each step.

MEMORY JOGGER

Think: same dollar step, not same percent step.

FORMULA / KEY MECHANICS

```
a_n = a_1 + (n-1)d where d = common difference The gap a_n - a_{n-1} is constant. This
creates straight-line growth. It is useful for linear schedules, not for most investment
growth paths.
```

REFERENCE TABLE

Term	Value if a_1=10, d=3	Interpretation
a_1	10	Starting level.
a_2	13	Add 3 once.
a_5	22	Add 3 four times.

WHY IT MATTERS IN QUANT RESEARCH

Arithmetic sequences appear in cash deposits, fixed withdrawals, laddered position sizes, price grid levels, and deterministic scenario assumptions.

SIMPLE MARKET / TRADING EXAMPLE

Depositing $200 each month creates a cash-flow sequence: 200, 200, 200, ... A price target grid could be 100, 102, 104, 106.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Use arithmetic schedules for planned cash flows or parameter grids. Do not use them as a default model for wealth compounding.

CONFUSIONS TO AVOID

A fixed dollar increase and a fixed percent return are different. A stock rising $1 every day has a shrinking percent return as price increases.

RELATED TERMS / QUICK RECALL

Linear growth, common difference, cash-flow schedule, grid search.

Geometric Sequences

ONE-SENTENCE MEANING

A geometric sequence changes by multiplying by the same fixed factor each step.

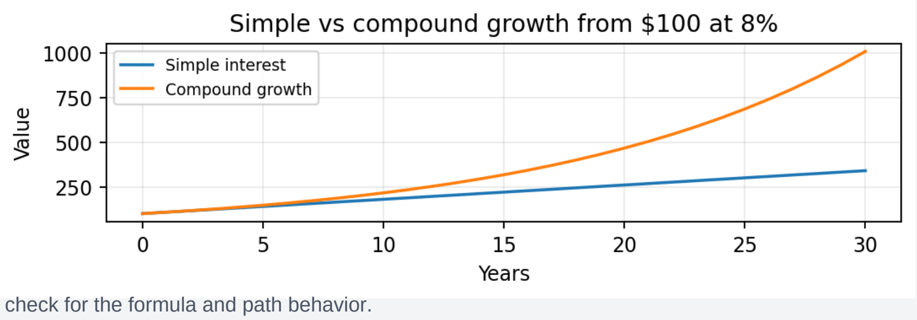
MEMORY JOGGER

Think: same percent step, not same dollar step.

FORMULA / KEY MECHANICS

```
a_n = a_1 x q^(n-1) q = common ratio growth factor g = 1 + r The ratio a_n / a_{n-1} is constant. If the ratio is greater than 1, the sequence grows exponentially. If it is between 0 and 1, it decays.
```

VISUAL / DIAGRAM



Use the visual as a quick check for the formula and path behavior.

WHY IT MATTERS IN QUANT RESEARCH

Financial compounding is geometric because each period return multiplies existing wealth. Volatility, fees, leverage, and reinvestment all act through multipliers.

SIMPLE MARKET / TRADING EXAMPLE

A portfolio that gains 5% each year from $100 follows 100, 105, 110.25, 115.76, ... because each year multiplies by 1.05.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Represent wealth updates with growth factors. Store both returns r_t and growth factors $g_t = 1+r_t$ when building performance analytics.

CONFUSIONS TO AVOID

Do not add returns to get final wealth unless returns are tiny and the approximation is explicitly acceptable.

RELATED TERMS / QUICK RECALL

Compound growth, growth factor, exponential growth, CAGR.

CORE MATH / BEGINNER / SEQUENCE BASICS

Recursive Sequences

ONE-SENTENCE MEANING

A recursive sequence defines each new value using one or more earlier values.

MEMORY JOGGER

Think: the next value is generated by a rule that looks backward.

FORMULA / KEY MECHANICS

```
Initial value: a_0 Update rule: a_t = f(a_{t-1}, input_t) A recursive rule needs an initial value and an update equation. Many financial objects are recursive: wealth updates, moving averages, volatility filters, inventory, and risk limits.
```

REFERENCE TABLE

Recursive object	Previous state	New input
Equity curve	Prior equity W_{t-1}	Current return r_t
EWMA volatility	Prior estimate	New squared return
Inventory	Prior position	New order fill

WHY IT MATTERS IN QUANT RESEARCH

Backtests are recursive systems. Yesterday equity, position, signal, and cash often determine today equity and allowable trade size.

SIMPLE MARKET / TRADING EXAMPLE

If W_t = W_{t-1}(1+r_t), then wealth cannot be computed for t without knowing W_{t-1}.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Design stateful components carefully. A recursive calculation must specify initialization, update order, and whether the value is available before or after execution.

CONFUSIONS TO AVOID

Recursive formulas are easy places to introduce look-ahead bias by updating with information that would not have been known yet.

RELATED TERMS / QUICK RECALL

State, update rule, recurrence relation, backtest engine.

CORE MATH / BEGINNER / SEQUENCE BASICS

Difference Equations and Update Rules

ONE-SENTENCE MEANING

A difference equation describes how a sequence changes from one step to the next.

MEMORY JOGGER

Think: it models the jump between now and next.

FORMULA / KEY MECHANICS

```
Delta a_t = a_t - a_{t-1} a_t = a_{t-1} + Delta a_t The first difference is Delta a_t = a_t - a_{t-1}. For prices, the raw difference is a dollar move; for returns, the percent difference is usually more comparable across assets.
```

REFERENCE TABLE

Update type	Equation	Meaning
Additive	x_t = x_{t-1} + d_t	Cash flows, differences.
Multiplicative	W_t = W_{t-1} x g_t	Compounded wealth.
Blended	m_t = alpha x x_t + (1-alpha)m_{t-1}	EWMA filters.

WHY IT MATTERS IN QUANT RESEARCH

Difference equations are the discrete-time version of dynamic systems. They show up in PnL, inventory, cash, position sizing, and factor decay.

SIMPLE MARKET / TRADING EXAMPLE

Cash after a trade can be written as cash_t = cash_{t-1} - shares_t x fill_price_t - costs_t.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

In implementation, write explicit update steps: compute signal, size position, execute, update cash, update equity, log state. This is cleaner than hiding everything in one expression.

CONFUSIONS TO AVOID

Do not mix price differences with returns. A $1 move in a $10 stock is not equivalent to a $1 move in a $500 stock.

RELATED TERMS / QUICK RECALL

Delta, state transition, discrete dynamics, simulation loop.

CORE MATH / BEGINNER / SEQUENCE BASICS

Monotonicity and Boundedness

ONE-SENTENCE MEANING

A monotonic sequence only moves one direction; a bounded sequence stays within fixed upper or lower limits.

MEMORY JOGGER

Think: monotonic means one-way; bounded means fenced in.

FORMULA / KEY MECHANICS

```
Increasing: a_{t+1} >= a_t Bounded above: a_t <= U for all t A sequence can be increasing, decreasing, nondecreasing, or nonincreasing. Boundedness asks whether values can run away to infinity or below a floor.
```

REFERENCE TABLE

Sequence	Monotonic?	Bounded?
Running max equity	Nondecreasing	Usually not bounded above.
Drawdown	No	Bounded between -1 and 0 in simple unlevered case.
Signal in [-1,1]	Maybe	Bounded by construction.

WHY IT MATTERS IN QUANT RESEARCH

Risk controls often create bounded sequences: position caps, leverage caps, stop-out rules, and exposure limits. Some performance metrics should be bounded by design.

SIMPLE MARKET / TRADING EXAMPLE

Cumulative maximum equity is nondecreasing. Drawdown is bounded above by 0 and below by -100% for an unlevered long-only portfolio.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Use bounds as validation checks. If a calculated drawdown is positive or less than -100% in an unlevered setting, your formula or data has a problem.

CONFUSIONS TO AVOID

A noisy upward trend is not monotonic. It can have a positive slope while still moving down at many individual steps.

RELATED TERMS / QUICK RECALL

Trend, bounds, cumulative max, validation rules.

CORE MATH / BEGINNER / SEQUENCE BASICS

Limits and Convergence

ONE-SENTENCE MEANING

A sequence converges when its terms approach a stable value as the index grows.

MEMORY JOGGER

Think: the values settle down toward a target.

FORMULA / KEY MECHANICS

```
lim_{n -> infinity} a_n = L means a_n gets arbitrarily close to L as n grows Convergence does not require hitting the final value exactly. It means the distance to the limiting value becomes arbitrarily small as n gets large.
```

REFERENCE TABLE

Object	Possible limit question	Research use
Sample mean	Does it stabilize?	Estimate expected return.
Monte Carlo average	Does simulation error shrink?	Validate scenario engine.
Optimization score	Does best parameter region persist?	Check robustness.

WHY IT MATTERS IN QUANT RESEARCH

Convergence matters when estimating averages, risk, expected return, simulation results, optimization parameters, and long-run growth rates.

SIMPLE MARKET / TRADING EXAMPLE

A sample mean computed on more and more observations may stabilize. A strategy estimate that keeps changing wildly with more data is not converging in practice.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Track metric stability as sample size grows. Plot expanding Sharpe, mean return, volatility, and drawdown to see whether conclusions stabilize.

CONFUSIONS TO AVOID

A sequence can look stable for a short sample and still fail out of sample. Convergence is a mathematical idea; empirical stability is a research diagnostic.

RELATED TERMS / QUICK RECALL

Limit, stability, estimator, long-run behavior.

CORE MATH / BEGINNER / SEQUENCE BASICS

Rate of Convergence

ONE-SENTENCE MEANING

Rate of convergence describes how fast a sequence approaches its limiting value.

MEMORY JOGGER

Think: settling slowly versus snapping into place.

FORMULA / KEY MECHANICS

```
Typical estimation error often shrinks like 1 / sqrt(n), not like 1 / n Two sequences can
converge to the same value but at different speeds. In estimation, speed controls how much
data you need before an estimate becomes usable.
```

REFERENCE TABLE

Estimator choice	Fast reaction	Slow reaction
Short lookback	Adapts quickly; noisy	Not applicable.
Long lookback	Slow to adapt	Stable but stale.
EWMA	Controlled by decay parameter	Recent data weighted more.

WHY IT MATTERS IN QUANT RESEARCH

Quant research often compares fast-reacting but noisy estimates with slow-reacting but stable estimates. Lookback length is a convergence-speed choice.

SIMPLE MARKET / TRADING EXAMPLE

A 5-day average reacts quickly to a volatility spike. A 252-day average converges more slowly but is less jumpy.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Monitor convergence using expanding plots and rolling diagnostics. If a metric only becomes stable after thousands of observations, short backtests are weak evidence.

CONFUSIONS TO AVOID

Fast convergence is not always better; it can mean overreacting to noise. Slow convergence can miss regime shifts.

RELATED TERMS / QUICK RECALL

Estimator stability, sample size, lookback, regime shift.

CORE MATH / BEGINNER / SERIES BASICS

Series as Summed Sequences

ONE-SENTENCE MEANING

A series is the sum of terms from a sequence.

MEMORY JOGGER

Think: sequence lists; series adds.

FORMULA / KEY MECHANICS

$S_n = a_1 + a_2 + ... + a_n = \sum_{i=1}^{n} a_i$ Given terms $a_1, a_2, ...$, the series is $a_1 + a_2 + ...$. A finite series has a fixed number of terms; an infinite series asks what happens as the number of terms grows without bound.

REFERENCE TABLE

Sequence	Series from it	Finance example
Trade costs	Cumulative costs	Total slippage paid.
Cash flows	Cumulative cash flow	Net deposits/withdrawals.
Log returns	Cumulative log return	Additive growth measure.

WHY IT MATTERS IN QUANT RESEARCH

Series appear in cumulative PnL, cumulative cash flows, discounted values, total transaction costs, rolling sums, and approximation formulas.

SIMPLE MARKET / TRADING EXAMPLE

Daily strategy PnL values form a sequence. The running account profit is a series of those PnL terms.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

When implementing analytics, name variables carefully: returns_t is a sequence; cumulative_return_t is a compounded path; cashflow_sum_t is a running series.

CONFUSIONS TO AVOID

Do not call every cumulative result a series. Compounded return is a product of growth factors, not a simple sum of simple returns.

RELATED TERMS / QUICK RECALL

Partial sum, cumulative sum, aggregation, PnL.

CORE MATH / BEGINNER / SERIES BASICS

Partial Sums

ONE-SENTENCE MEANING

A partial sum is the running total after adding only the first n terms of a series.

MEMORY JOGGER

Think: the scoreboard after each step.

FORMULA / KEY MECHANICS

```
S_n = S_{n-1} + a_n, with S_0 = 0 Partial sums create a new sequence: S_1, S_2, S_3, ... .
Studying that sequence shows whether totals stabilize, drift, or explode.
```

REFERENCE TABLE

n	a_n	S_n
1	5	5
2	-2	3
3	4	7
4	-6	1

WHY IT MATTERS IN QUANT RESEARCH

Partial sums are used for cumulative PnL, cumulative log return, cumulative volume, rolling sums, and expanding-window metrics.

SIMPLE MARKET / TRADING EXAMPLE

If daily PnL is [5, -2, 4], partial sums are [5, 3, 7]. That running path is more informative than final PnL alone.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Keep both period values and cumulative values in reports. Final totals hide when profit arrived and whether the path was survivable.

CONFUSIONS TO AVOID

A high final partial sum can still involve a deep drawdown. Do not evaluate a strategy only by the last cumulative value.

RELATED TERMS / QUICK RECALL

Cumsum, running total, equity path, trajectory.

Arithmetic Series

ONE-SENTENCE MEANING

An arithmetic series is the sum of an arithmetic sequence.

MEMORY JOGGER

Think: adding evenly spaced values.

FORMULA / KEY MECHANICS

`S_n = n(a_1 + a_n)/2 = n/2 x [2a_1 + (n-1)d] Because the terms rise or fall by a constant difference, the average of the first and last term gives the average term value. Multiply by n to get the sum.`

REFERENCE TABLE

Use case	Terms	Series meaning
Order ladder	100, 200, 300 shares	Total shares across ladder.
Savings plan	$100, $125, $150	Total planned contributions.
Parameter grid	5,10,15,20	Total grid points if paired with other axes.

WHY IT MATTERS IN QUANT RESEARCH

Arithmetic series are useful for laddered orders, linear savings schedules, planned dollar deposits, and evenly spaced parameter grids.

SIMPLE MARKET / TRADING EXAMPLE

If you add 100, 150, 200, 250, the sum is 4 x (100+250)/2 = 700.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Use arithmetic series to estimate total capital deployed across a ladder or total notional in a deterministic scaling plan.

CONFUSIONS TO AVOID

Do not use an arithmetic series to sum compounded wealth levels as if they were returns. Wealth levels and cash flows are different objects.

RELATED TERMS / QUICK RECALL

Common difference, ladder, linear schedule, total allocation.

CORE MATH / BEGINNER / SERIES BASICS

Geometric Series

ONE-SENTENCE MEANING

A geometric series is the sum of terms that each multiply by a constant ratio.

MEMORY JOGGER

Think: adding values that shrink or grow by a fixed percent.

FORMULA / KEY MECHANICS

```
S_n = a(1-q^n)/(1-q), q != 1 Infinite: S = a/(1-q) if |q| < 1 When the ratio is between -1
and 1, the infinite sum can converge. This is the math behind many discounted cash-flow and
decay-weight formulas.
```

REFERENCE TABLE

q value	Behavior	Finance interpretation
$0 < q < 1$	Terms shrink	Discounted future values.
$q = 1$	Terms stay same	Sum grows linearly.
$q > 1$	Terms grow	No finite infinite sum.

WHY IT MATTERS IN QUANT RESEARCH

Discounting, decay weights, present value, and some infinite-horizon models rely on geometric series.

SIMPLE MARKET / TRADING EXAMPLE

A payment worth 100 today, discounted by 0.95 each period, forms 100 + 95 + 90.25 + ...

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Use geometric series when future terms are scaled by a constant discount factor or decay ratio. This keeps long horizons mathematically manageable.

CONFUSIONS TO AVOID

If the ratio is >= 1 in absolute value, the infinite series does not converge to a finite value.

RELATED TERMS / QUICK RECALL

Discount factor, decay, present value, infinite horizon.

CORE MATH / BEGINNER / SERIES BASICS

Infinite Geometric Series

ONE-SENTENCE MEANING

An infinite geometric series adds endlessly many geometrically shrinking terms and can still have a finite value.

MEMORY JOGGER

Think: infinitely many smaller and smaller pieces can fit inside a finite box.

FORMULA / KEY MECHANICS

```
a + aq + aq^2 + ... = a/(1-q), only when |q| < 1 The key condition is |q| < 1. Each
additional term contributes less than the last, so the partial sums approach a limit.
```

REFERENCE TABLE

Question	Check	Interpretation
Does it converge?	Is $|q| < 1$?	If yes, finite value.
How much is left after n terms?	Remainder = aq^n/(1-q)	Tail risk of truncation.
Is q a discount factor?	q = 1/(1+r)	Positive r gives $q < 1$.

WHY IT MATTERS IN QUANT RESEARCH

This is central to perpetuities, discounted infinite cash flows, long-memory approximations, and decay-weighted indicators.

SIMPLE MARKET / TRADING EXAMPLE

A perpetual $10 payment discounted at 5% has present value 10 / 0.05 = 200 under the simple perpetuity formula.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

When using infinite-horizon formulas, document the discount rate and confirm that the effective ratio is below 1.

CONFUSIONS TO AVOID

Infinite does not mean infinite value. But the convergence condition must hold; otherwise the formula is invalid.

RELATED TERMS / QUICK RECALL

Perpetuity, discounting, convergence, horizon approximation.

CORE MATH / BEGINNER / SERIES BASICS

Telescoping Series

ONE-SENTENCE MEANING

A telescoping series cancels most middle terms, leaving only a small number of boundary terms.

MEMORY JOGGER

Think: the middle collapses and the endpoints remain.

FORMULA / KEY MECHANICS

```
sum_{t=1..T} (b_t - b_{t-1}) = b_T - b_0 A term like b_t - b_{t-1} creates cancellation when summed over time. This is common when moving from period changes to total change.
```

REFERENCE TABLE

Object	Telescopes?	Reason
Price differences	Yes	Middle prices cancel.
Simple returns	No	Percent changes compound.
Log returns	Yes in log space	Log ratios add.

WHY IT MATTERS IN QUANT RESEARCH

Returns are not always telescoping, but price differences are. Cumulative dollar price change telescopes from first price to last price.

SIMPLE MARKET / TRADING EXAMPLE

If price changes are $P_t - P_{t-1}$, then summing them from t=1 to T gives $P_T - P_0$.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Use telescoping logic to sanity-check cumulative differences, inventory changes, and cash balance updates.

CONFUSIONS TO AVOID

Simple returns do not telescope by addition. Log prices do: $\sum \log(P_t/P_{t-1}) = \log(P_T/P_0)$.

RELATED TERMS / QUICK RECALL

Cancellation, boundary terms, price differences, log returns.

CORE MATH / BEGINNER / SERIES BASICS

Convergence Tests: Practical Map

ONE-SENTENCE MEANING

Convergence tests are rules for deciding whether an infinite series settles to a finite value.

MEMORY JOGGER

Think: a checklist for whether endless adding behaves.

FORMULA / KEY MECHANICS

```
If terms do not go to 0, the series diverges. If they do go to 0, more testing is needed.
Beginner quant work rarely requires proving convergence formally, but the intuition matters:
shrinking terms alone are not always enough; they must shrink fast enough.
```

REFERENCE TABLE

Series type	Rule of thumb	Finance analog
Geometric	Converges if $\lvert q\rvert < 1$	Discounting and decay.
Harmonic-like	Can diverge slowly	Long tails can matter.
Fast decay	Usually safe to truncate	Short-memory filters.

WHY IT MATTERS IN QUANT RESEARCH

Convergence logic appears in discounted values, infinite horizons, approximation error, decay weights, and simulation truncation.

SIMPLE MARKET / TRADING EXAMPLE

A discount factor of 0.99 converges, but slowly. Truncating after 20 terms may miss meaningful tail value.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

When approximating infinite sums, compute or bound the remaining tail. Do not assume a fixed horizon is long enough.

CONFUSIONS TO AVOID

A term going to zero is necessary but not sufficient. The harmonic series terms 1/n go to zero, but the sum diverges.

RELATED TERMS / QUICK RECALL

Ratio test, comparison, tail error, truncation.

CORE MATH / BEGINNER / SERIES BASICS

Power Series and Local Approximation

ONE-SENTENCE MEANING

A power series expresses a function as a sum of powers of a variable around a point.

MEMORY JOGGER

Think: approximate a curved function with layered polynomial pieces.

FORMULA / KEY MECHANICS

`log(1+r) = r - r^2/2 + r^3/3 - ... for |r| < 1 Power series let you replace a complicated function with a polynomial approximation near a chosen point. The approximation is local: it works best near the expansion point.`

REFERENCE TABLE

Approximation	Works when	Risk
log(1+r) ~ r	r is small	Overstates after large negative returns.
exp(x) ~ 1+x	x is small	Misses curvature.
Polynomial expansion	Near expansion point	Bad extrapolation.

WHY IT MATTERS IN QUANT RESEARCH

Quant models use approximations for log returns, option sensitivities, risk expansions, and numerical methods.

SIMPLE MARKET / TRADING EXAMPLE

For small r, log(1+r) is close to r. Adding more terms improves approximation for small moves but can fail for large moves.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Use approximation formulas only within their valid range. In code, prefer exact functions like log1p(r) when available.

CONFUSIONS TO AVOID

A good local approximation can become bad in stress regimes. Always test approximation error across realistic input ranges.

RELATED TERMS / QUICK RECALL

Taylor series, log approximation, numerical stability, local model.

CORE MATH / BEGINNER / SERIES BASICS

Series Expansion as Model Thinking

ONE-SENTENCE MEANING

Series expansion is the habit of explaining a complex quantity as a base term plus correction terms.

MEMORY JOGGER

Think: start simple, then add layers of realism.

FORMULA / KEY MECHANICS

```
Observed performance = base model + correction_1 + correction_2 + ... + residual A first term gives a rough model. Later terms capture curvature, interaction, decay, or higher-order effects. This is a useful mental model even when you are not doing formal calculus.
```

REFERENCE TABLE

Layer	Question	Example
Base term	What happens in frictionless math?	Raw signal return.
Correction	What real-world force reduces it?	Fees and slippage.
Residual	What remains unexplained?	Noise or model error.

WHY IT MATTERS IN QUANT RESEARCH

Performance attribution often works this way: gross return, minus costs, minus slippage, minus borrow, minus fees, plus financing effects.

SIMPLE MARKET / TRADING EXAMPLE

A naive backtest says +12%. A correction series says +12% - 2% costs - 1% slippage - 0.5% borrow = +8.5% before taxes.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Build research reports with layered adjustments. Show how each correction term changes the result instead of only showing final performance.

CONFUSIONS TO AVOID

Correction terms are not decorative. Missing one important term can flip a strategy from profitable to untradable.

RELATED TERMS / QUICK RECALL

Approximation, attribution, decomposition, correction terms.

CORE MATH / BEGINNER / FINANCIAL GROWTH

Simple Interest

ONE-SENTENCE MEANING

Simple interest grows by adding interest only on the original principal.

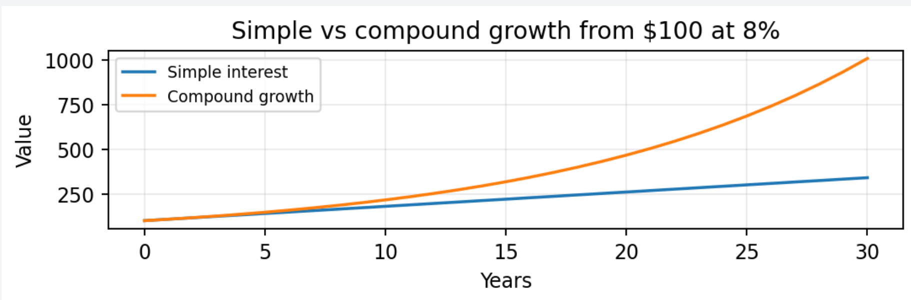
MEMORY JOGGER

Think: interest ignores previous interest.

FORMULA / KEY MECHANICS

`FV = P x (1 + r x n) The dollar interest each period is constant when rate and principal are fixed. This creates linear growth rather than exponential growth.`

VISUAL / DIAGRAM



Use the visual as a quick check for the formula and path behavior.

WHY IT MATTERS IN QUANT RESEARCH

Simple interest is less common for investment wealth but useful as a baseline, loan simplification, and contrast against compounding.

SIMPLE MARKET / TRADING EXAMPLE

$1,000 at 10% simple interest earns $100 per year. After 3 years, value is $1,300, not $1,331.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Use simple interest examples to separate additive cash-flow thinking from multiplicative growth thinking.

CONFUSIONS TO AVOID

Do not use simple interest for reinvested strategies, ETFs, equity curves, or portfolio wealth unless the setup explicitly prevents reinvestment.

RELATED TERMS / QUICK RECALL

Principal, interest rate, linear growth, non-compounded return.

CORE MATH / BEGINNER / FINANCIAL GROWTH

Compound Interest

ONE-SENTENCE MEANING

Compound interest grows by earning returns on previous returns.

MEMORY JOGGER

Think: the base being multiplied keeps getting larger.

FORMULA / KEY MECHANICS

FV = P x (1+r)^n W_t = W_{t-1} x (1+r_t) Compounding is geometric. Each period applies a growth factor to the full current value, not just to the starting principal.

REFERENCE TABLE

Path	Arithmetic sum	Final wealth from $100
+10%, +10%	+20%	$121
+50%, -50%	0%	$75
-20%, +25%	+5%	$100

WHY IT MATTERS IN QUANT RESEARCH

Every reinvested strategy, buy-and-hold path, equity curve, and multi-period return calculation is a compounding problem.

SIMPLE MARKET / TRADING EXAMPLE

$1,000 at 10% compounded annually becomes 1000 x 1.1^3 = $1,331 after 3 years.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Backtest engines should update equity recursively: W_t = W_{t-1}(1+r_t), after costs and financing are applied.

CONFUSIONS TO AVOID

A +50% return followed by -50% return does not break even; wealth goes 100 -> 150 -> 75.

RELATED TERMS / QUICK RECALL

Growth factor, reinvestment, CAGR, equity curve.

CORE MATH / BEGINNER / FINANCIAL GROWTH

Discrete Compounding Intervals

ONE-SENTENCE MEANING

Discrete compounding applies interest or return a fixed number of times per year.

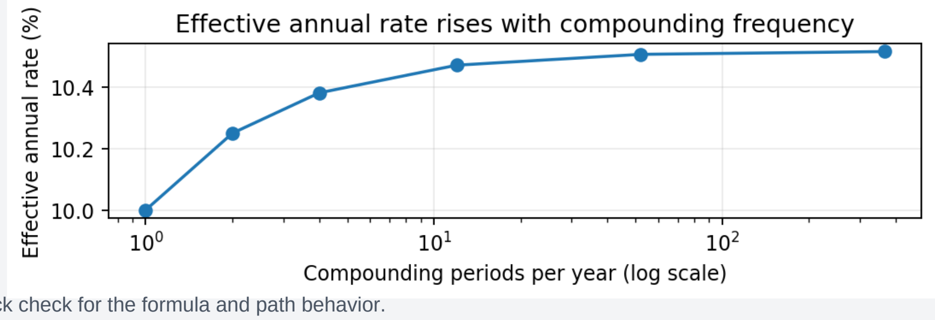
MEMORY JOGGER

Think: compounding has a clock.

FORMULA / KEY MECHANICS

`FV = P x (1 + r/m)^(m n) EAR = (1 + r/m)^m - 1 More frequent compounding increases effective return for the same quoted nominal rate, but the gain diminishes as frequency rises.`

VISUAL / DIAGRAM



Use the visual as a quick check for the formula and path behavior.

WHY IT MATTERS IN QUANT RESEARCH

Rates must be compared on the same compounding basis. Daily, monthly, quarterly, and annual rates are not directly interchangeable without conversion.

SIMPLE MARKET / TRADING EXAMPLE

A 12% nominal rate compounded monthly has monthly rate 1% and effective annual rate (1.01)^12 - 1 = 12.68%.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Normalize rates before comparing strategy returns, financing rates, margin rates, yields, or borrowing costs.

CONFUSIONS TO AVOID

Do not compare a nominal APR to a compounded backtest CAGR as if they are the same type of number.

RELATED TERMS / QUICK RECALL

APR, APY, effective annual rate, compounding frequency.

CORE MATH / BEGINNER / FINANCIAL GROWTH

Continuous Compounding

ONE-SENTENCE MEANING

Continuous compounding is the limiting case where compounding happens infinitely often.

MEMORY JOGGER

Think: smooth compounding with e as the growth engine.

FORMULA / KEY MECHANICS

```
FV = P x e^(r t) continuously compounded return = ln(FV/P) The formula uses exp(r t).
Continuous compounding is mathematically convenient because log returns and continuously
compounded rates add cleanly over time.
```

REFERENCE TABLE

Rate language	Formula	Use
Simple rate	P(1+rt)	Linear baseline.
Discrete compound	P(1+r/m)^(mt)	Quoted intervals.
Continuous	P e^(rt)	Log-return math.

WHY IT MATTERS IN QUANT RESEARCH

Used in derivatives, risk models, yield conversion, log return analysis, and continuous-time finance approximations.

SIMPLE MARKET / TRADING EXAMPLE

$1,000 at continuously compounded 10% for 3 years is 1000 x exp(0.30) = about $1,349.86.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Use exp and log carefully. In Python, prefer numpy.exp and numpy.log1p for numerical stability when returns are small.

CONFUSIONS TO AVOID

Continuous compounding is not automatically more realistic; it is often a modeling convention.

RELATED TERMS / QUICK RECALL

e, log return, continuously compounded rate, exponential growth.

CORE MATH / BEGINNER / FINANCIAL GROWTH

Effective Annual Rate

ONE-SENTENCE MEANING

Effective annual rate is the true one-year growth rate after compounding is included.

MEMORY JOGGER

Think: translate the quote into actual yearly growth.

FORMULA / KEY MECHANICS

`EAR = (1 + quoted_rate / m)^m - 1 EAR makes rates comparable by putting them on the same annual compounded basis. A nominal rate can understate the actual annual multiplier when compounding occurs within the year.`

REFERENCE TABLE

Quoted compounding	m	EAR for 10% nominal
Annual	1	10.00%
Quarterly	4	10.38%
Monthly	12	10.47%
Daily	365	10.52%

WHY IT MATTERS IN QUANT RESEARCH

Use EAR to compare savings rates, margin rates, borrowing costs, yield assumptions, and performance targets.

SIMPLE MARKET / TRADING EXAMPLE

A nominal 12% compounded monthly produces EAR = (1+0.12/12)^12 - 1 = 12.68%.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

In dashboards, store both quoted rate and effective converted rate. The converted value should be used in apples-to-apples comparisons.

CONFUSIONS TO AVOID

APR and APY are often confused. APR is usually a quoted annualized rate; APY/EAR includes compounding.

RELATED TERMS / QUICK RECALL

APR, APY, annualization, rate conversion.

CORE MATH / BEGINNER / FINANCIAL GROWTH

Growth Factor Notation

ONE-SENTENCE MEANING

A growth factor is the multiplier 1+r applied to wealth for one period.

MEMORY JOGGER

Think: return as a multiplier, not just a percent label.

FORMULA / KEY MECHANICS

```
g_t = 1 + r_t W_T = W_0 x product_{t=1..T} g_t If r_t is return, then g_t = 1+r_t. Positive
returns create g_t > 1; negative returns create 0 < g_t < 1; a -100% return creates g_t = 0.
```

REFERENCE TABLE

Return r_t	Growth factor g_t	Effect on $100
+20%	1.20	$120
0%	1.00	$100
-30%	0.70	$70
-100%	0.00	$0

WHY IT MATTERS IN QUANT RESEARCH

Growth factors make compounding clear because multi-period wealth is a product of factors.

SIMPLE MARKET / TRADING EXAMPLE

Returns [+10%, -5%] become growth factors [1.10, 0.95]. Total multiplier is 1.10 x 0.95 = 1.045.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

In code, calculate growth = 1 + returns, then cumulative wealth = initial * growth.cumprod(). This avoids accidental additive thinking.

CONFUSIONS TO AVOID

A return less than -100% is invalid for ordinary unlevered wealth, but leveraged strategies can create equity below zero in model assumptions if not constrained.

RELATED TERMS / QUICK RECALL

Multiplier, gross return, cumprod, wealth index.

CORE MATH / BEGINNER / FINANCIAL GROWTH

Cumulative Return as Product

ONE-SENTENCE MEANING

Cumulative return over many periods comes from multiplying growth factors and subtracting 1.

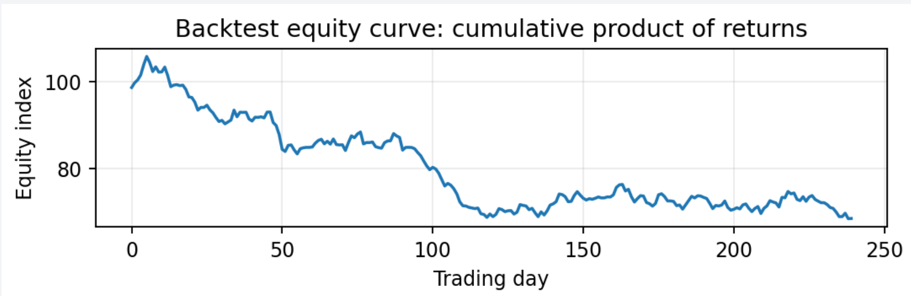
MEMORY JOGGER

Think: multiply the path, then convert back to a percent.

FORMULA / KEY MECHANICS

```
R_{0:T} = product_{t=1..T}(1+r_t) - 1 Simple returns add poorly across time. The correct
cumulative simple return is product(1+r_t)-1.
```

VISUAL / DIAGRAM



Use the visual as a quick check for the formula and path behavior.

WHY IT MATTERS IN QUANT RESEARCH

This is the core equation behind equity curves, performance reports, and portfolio growth analytics.

SIMPLE MARKET / TRADING EXAMPLE

If returns are +10%, +5%, -3%, cumulative return is 1.10 x 1.05 x 0.97 - 1 = 12.04%.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Use vectorized cumulative products for return streams. Always include costs, slippage, and financing before compounding if they affect realizable wealth.

CONFUSIONS TO AVOID

Adding period returns is only an approximation for small returns and short horizons. It fails badly for volatile paths.

RELATED TERMS / QUICK RECALL

Cumulative return, growth factor, equity curve, compounding.

Log Returns as Additive Growth Series

ONE-SENTENCE MEANING

A log return is the natural log of a growth factor, making multi-period growth additive in log space.

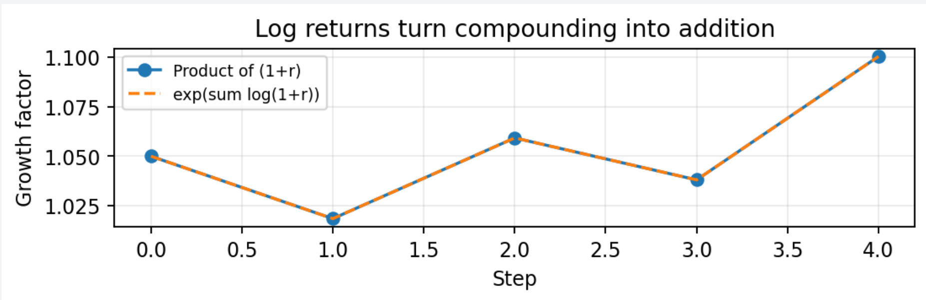
MEMORY JOGGER

Think: compounding becomes summing after taking logs.

FORMULA / KEY MECHANICS

```
ell_t = ln(P_t/P_{t-1}) = ln(1+r_t) sum ell_t = ln(P_T/P_0) Log returns are defined as
ln(P_t/P_{t-1}) = ln(1+r_t). Summing log returns gives the log of the total growth
multiplier.
```

VISUAL / DIAGRAM



Use the visual as a quick check for the formula and path behavior.

WHY IT MATTERS IN QUANT RESEARCH

Log returns simplify time aggregation, statistical modeling, and continuous compounding links. They are common in quantitative research.

SIMPLE MARKET / TRADING EXAMPLE

If log returns sum to 0.20, the total growth factor is exp(0.20) = 1.221, so cumulative return is 22.1%.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

In Python, use np.log(price).diff() or np.log1p(simple_returns). Use exp(cumsum(log_returns)) for a wealth multiplier.

CONFUSIONS TO AVOID

Log returns are not the same as simple returns for large moves. Do not present log return as investor percent gain without converting.

RELATED TERMS / QUICK RECALL

Log return, continuous compounding, additive series, telescoping.

CORE MATH / BEGINNER / FINANCIAL GROWTH

Arithmetic Mean vs Geometric Mean

ONE-SENTENCE MEANING

The arithmetic mean averages period returns; the geometric mean measures the constant compounded return that matches final wealth.

MEMORY JOGGER

Think: arithmetic mean describes average period; geometric mean describes actual growth rate.

FORMULA / KEY MECHANICS

```
Arithmetic mean = (1/n) sum r_t Geometric mean = [product(1+r_t)]^(1/n) - 1 The arithmetic mean is usually higher when returns are volatile because it ignores volatility drag. The geometric mean embeds compounding.
```

REFERENCE TABLE

Metric	Answers	Weakness
Arithmetic mean	Average period return	Ignores compounding drag.
Geometric mean	Constant compounded rate	Less direct for one-period expectation.
CAGR	Annualized geometric growth	Sensitive to start/end dates.

WHY IT MATTERS IN QUANT RESEARCH

Use arithmetic mean for expected one-period return assumptions; use geometric mean or CAGR for multi-period realized growth.

SIMPLE MARKET / TRADING EXAMPLE

Returns +50% and -50% have arithmetic mean 0%, but $100 becomes $75, so the geometric result is negative.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Performance reports should show CAGR for realized multi-period growth and disclose volatility separately.

CONFUSIONS TO AVOID

A strategy can have a positive average return and still lose money after compounding if losses are large enough.

RELATED TERMS / QUICK RECALL

Average return, CAGR, volatility drag, realized growth.

CORE MATH / BEGINNER / FINANCIAL GROWTH

Volatility Drag

ONE-SENTENCE MEANING

Volatility drag is the loss in compounded growth caused by returns bouncing around.

MEMORY JOGGER

Think: losses hurt more than equal-sized gains help.

FORMULA / KEY MECHANICS

```
Approximation: geometric return ~ arithmetic return - 0.5 x variance A -20% loss needs +25%
to recover. Because compounding is multiplicative, variability lowers geometric growth
relative to arithmetic average.
```

REFERENCE TABLE

Loss	Gain needed to recover	Reason
-10%	+11.1%	90 to 100.
-25%	+33.3%	75 to 100.
-50%	+100%	50 to 100.

WHY IT MATTERS IN QUANT RESEARCH

Volatility drag explains why smoother strategies can outperform choppy strategies with similar average returns.

SIMPLE MARKET / TRADING EXAMPLE

A path of +10%, -10% has average 0%, but final wealth is 100 x 1.1 x 0.9 = 99.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Evaluate strategies using geometric growth, drawdown, and volatility together. A high average return is not enough.

CONFUSIONS TO AVOID

Do not assume symmetric return percentages have symmetric wealth effects. Down moves require larger up moves to recover.

RELATED TERMS / QUICK RECALL

Geometric mean, variance, drawdown, risk-adjusted return.

Drawdown and Recovery Sequences

ONE-SENTENCE MEANING

Drawdown tracks the percent drop from a running peak in an equity sequence.

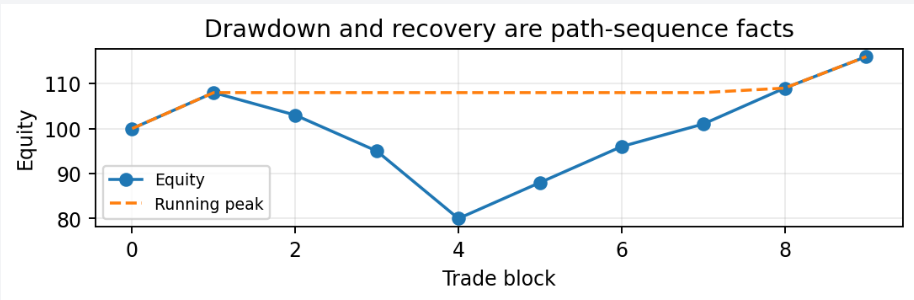
MEMORY JOGGER

Think: how far underwater the path is compared with its best prior point.

FORMULA / KEY MECHANICS

`Drawdown_t = W_t / max(W_0, ..., W_t) - 1 Drawdown is path-dependent. It requires the running maximum and current value at each step, not just final return.`

VISUAL / DIAGRAM



Use the visual as a quick check for the formula and path behavior.

WHY IT MATTERS IN QUANT RESEARCH

Drawdowns measure survivability, psychological pressure, margin stress, and capital impairment in strategy evaluation.

SIMPLE MARKET / TRADING EXAMPLE

If equity goes 100, 120, 90, the drawdown at 90 is 90/120 - 1 = -25%.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Backtest reports should compute running peak, drawdown series, max drawdown, drawdown duration, and recovery time.

CONFUSIONS TO AVOID

A strategy with high CAGR but massive drawdowns may be unusable. Final wealth does not reveal whether the investor could survive the path.

RELATED TERMS / QUICK RECALL

Running max, max drawdown, recovery, underwater curve.

CORE MATH / BEGINNER / FINANCIAL GROWTH

CAGR: Compound Annual Growth Rate

ONE-SENTENCE MEANING

CAGR is the constant annual compounded rate that turns starting value into ending value over a time span.

MEMORY JOGGER

Think: smooth annual rate equivalent to the full path.

FORMULA / KEY MECHANICS

```
CAGR = (Ending value / Starting value)^(1/years) - 1 CAGR ignores the shape of the path and only uses start value, end value, and elapsed years. It is useful but incomplete.
```

REFERENCE TABLE

Input	Value	Role
Start value	$10,000	Base wealth.
End value	$16,000	Final wealth.
Years	4	Annualization horizon.
CAGR	12.47%	Equivalent annual compound rate.

WHY IT MATTERS IN QUANT RESEARCH

CAGR is a standard performance headline for strategies, portfolios, benchmarks, and scenarios.

SIMPLE MARKET / TRADING EXAMPLE

If $10,000 becomes $16,000 in 4 years, CAGR = (16000/10000)^(1/4)-1 = 12.47%.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Use CAGR with max drawdown, volatility, Sharpe, turnover, and exposure. Never rely on CAGR alone.

CONFUSIONS TO AVOID

CAGR can be inflated by choosing a favorable start date. Always show the evaluation period clearly.

RELATED TERMS / QUICK RECALL

Annualization, geometric mean, performance reporting, endpoint bias.

CORE MATH / BEGINNER / FINANCIAL GROWTH

Present Value and Discounting

ONE-SENTENCE MEANING

Present value converts a future amount into today's value using a discount rate.

MEMORY JOGGER

Think: future money is divided by a growth factor.

FORMULA / KEY MECHANICS

```
PV = FV / (1+r)^n Discount factor = 1/(1+r)^n Discounting reverses compounding. Instead of multiplying forward by (1+r)^n, divide future value by (1+r)^n.
```

REFERENCE TABLE

Higher discount rate	PV effect	Interpretation
Low r	Higher present value	Future cash valued more.
High r	Lower present value	Future cash penalized more.
Longer n	Lower present value	More waiting.

WHY IT MATTERS IN QUANT RESEARCH

Used in valuation, cash-flow modeling, bond math, opportunity cost, and comparing returns across time.

SIMPLE MARKET / TRADING EXAMPLE

$1,000 received 3 years from now at a 5% discount rate is 1000/(1.05)^3 = $863.84 today.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Discount future cash flows before summing them. A cash flow today and a cash flow five years from now are not equivalent.

CONFUSIONS TO AVOID

The discount rate is an assumption, not a fact. Small changes can materially change present value.

RELATED TERMS / QUICK RECALL

Future value, discount factor, DCF, opportunity cost.

CORE MATH / BEGINNER / FINANCIAL GROWTH

Future Value of Cash Flows

ONE-SENTENCE MEANING

Future value accumulates one or more cash flows forward to a later date.

MEMORY JOGGER

Think: move every cash flow to the same future scoreboard.

FORMULA / KEY MECHANICS

FV_T = sum_{k=0..T} CF_k x (1+r)^(T-k) A cash flow at time k compounds for T-k periods before the final date. Different cash-flow dates get different exponents.

REFERENCE TABLE

Cash flow	Years compounded to T=3	Future value at 6%
$500 at t=0	3	$595.51
$500 at t=1	2	$561.80
$500 at t=3	0	$500.00

WHY IT MATTERS IN QUANT RESEARCH

Portfolio contributions, withdrawals, dividends, subscriptions, and rebalancing cash all require time-aware accumulation.

SIMPLE MARKET / TRADING EXAMPLE

$500 deposited today and $500 next year at 6% for 3 years becomes 500(1.06)^3 + 500(1.06)^2.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Record cash-flow dates explicitly. Performance systems must separate investment return from external contributions.

CONFUSIONS TO AVOID

Do not treat late deposits as if they were invested from day one. That overstates performance contribution.

RELATED TERMS / QUICK RECALL

Cash flow timing, money-weighted return, FV, contribution schedule.

CORE MATH / BEGINNER / FINANCIAL GROWTH

Annuities and Recurring Deposits

ONE-SENTENCE MEANING

An annuity is a sequence of equal cash flows paid or received at regular intervals.

MEMORY JOGGER

Think: the same payment repeated on a schedule.

FORMULA / KEY MECHANICS

```
FV of ordinary annuity = PMT x [((1+r)^n - 1) / r] Annuity formulas are geometric-series shortcuts for valuing repeated payments. Timing matters: payments at the beginning versus end of periods have different values.
```

REFERENCE TABLE

Annuity type	Payment timing	Value effect
Ordinary annuity	End of period	Standard formula.
Annuity due	Beginning of period	One extra period of growth.
Irregular deposits	Variable timing	Use explicit cash-flow loop.

WHY IT MATTERS IN QUANT RESEARCH

Used for recurring investments, loan payments, subscriptions, retirement contributions, and systematic withdrawal plans.

SIMPLE MARKET / TRADING EXAMPLE

Investing $200 monthly creates a repeated cash-flow sequence. Each deposit compounds for a different number of months.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

In research tools, model contributions separately from returns so you can distinguish saving behavior from strategy performance.

CONFUSIONS TO AVOID

An annuity formula assumes regular equal payments and constant rate. Real markets violate this, so simulations often beat closed-form formulas.

RELATED TERMS / QUICK RECALL

Recurring cash flow, contribution plan, loan payment, geometric series.

CORE MATH / BEGINNER / FINANCIAL GROWTH

Dollar-Cost Averaging as a Sequence

ONE-SENTENCE MEANING

Dollar-cost averaging invests a fixed dollar amount through time, creating a sequence of purchases at different prices.

MEMORY JOGGER

Think: same dollars, variable shares.

FORMULA / KEY MECHANICS

```
Shares bought_t = contribution_t / price_t Total shares = sum shares_t When price is low, the fixed contribution buys more units. When price is high, it buys fewer units. The final result depends on the entire price path.
```

REFERENCE TABLE

Price	Contribution	Shares bought
$10	$100	10.00
$20	$100	5.00
$25	$100	4.00

WHY IT MATTERS IN QUANT RESEARCH

DCA is a cash-flow and path-dependency problem, not just a return problem.

SIMPLE MARKET / TRADING EXAMPLE

Investing $100 at prices 10, 20, and 25 buys 10, 5, and 4 shares for 19 total shares.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

A backtest with contributions should track shares, cash, price, and contribution schedule explicitly.

CONFUSIONS TO AVOID

Do not compare a DCA path directly to a lump-sum path without specifying cash timing and opportunity cost.

RELATED TERMS / QUICK RECALL

Shares, contributions, average cost, path dependency.

CORE MATH / BEGINNER / FINANCIAL GROWTH

Inflation and Real Growth

ONE-SENTENCE MEANING

Real growth measures purchasing-power growth after inflation is removed.

MEMORY JOGGER

Think: nominal wealth minus the price-level erosion.

FORMULA / KEY MECHANICS

`1 + real return = (1 + nominal return) / (1 + inflation rate) Nominal return is what the account shows. Real return adjusts for inflation by comparing wealth growth to the inflation growth factor.`

REFERENCE TABLE

Nominal	Inflation	Exact real
7%	3%	3.88%
10%	8%	1.85%
2%	5%	-2.86%

WHY IT MATTERS IN QUANT RESEARCH

Long-horizon research should distinguish nominal performance from real performance because inflation compounds too.

SIMPLE MARKET / TRADING EXAMPLE

A 7% nominal return with 3% inflation is approximately 4% real, but exact real return is 1.07/1.03 - 1 = 3.88%.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Use real growth in long-term projections, spending models, and comparisons across high-inflation regimes.

CONFUSIONS TO AVOID

Do not subtract inflation for exact multi-period calculations when compounding matters. Use growth-factor division.

RELATED TERMS / QUICK RECALL

Nominal return, real return, purchasing power, inflation adjustment.

Fees, Taxes, and Slippage Compound Too

ONE-SENTENCE MEANING

Small recurring frictions can create large long-run wealth differences because they compound through time.

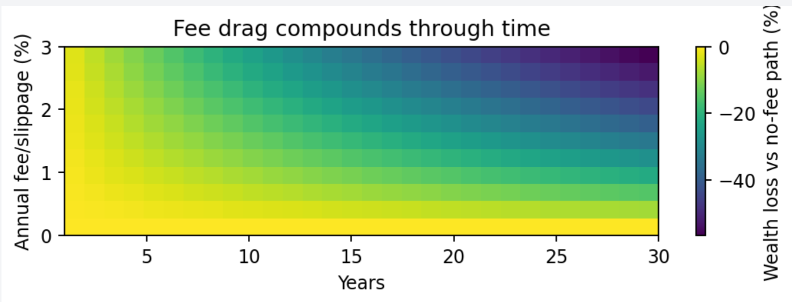
MEMORY JOGGER

Think: a tiny leak gets multiplied across the whole path.

FORMULA / KEY MECHANICS

`Net growth factor_t = 1 + gross_return_t - cost_t - fee_t - financing_t Fees, taxes, spreads, commissions, slippage, borrow, and financing costs reduce the growth factor before compounding.`

VISUAL / DIAGRAM



Use the visual as a quick check for the formula and path behavior.

WHY IT MATTERS IN QUANT RESEARCH

Many strategies look profitable before costs and weak after costs. Cost modeling is not optional.

SIMPLE MARKET / TRADING EXAMPLE

A 1% annual drag over 30 years can remove a large share of terminal wealth compared with the no-cost path.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Apply costs at the trade or period level before computing cumulative performance. Report gross and net results side by side.

CONFUSIONS TO AVOID

Do not subtract one final lump cost after compounding if the cost actually occurs throughout the path.

RELATED TERMS / QUICK RECALL

Transaction costs, net return, turnover, fee drag.

CORE MATH / BEGINNER / FINANCIAL GROWTH

Leverage and Margin Growth Paths

ONE-SENTENCE MEANING

Leverage magnifies the return sequence by increasing exposure relative to equity.

MEMORY JOGGER

Think: bigger multiplier, bigger damage when wrong.

FORMULA / KEY MECHANICS

```
Levered return_t ~ L x asset_return_t - financing_t - costs_t A leverage factor L turns an asset return r_t into roughly L x r_t before financing and constraints. Losses compound against equity and can trigger margin limits.
```

REFERENCE TABLE

Asset move	1x equity effect	3x approximate effect
+10%	+10%	+30%
-10%	-10%	-30%
-33.3%	-33.3%	about -100%

WHY IT MATTERS IN QUANT RESEARCH

Leverage makes path dependency more severe. Volatility drag, drawdown, and ruin risk rise quickly.

SIMPLE MARKET / TRADING EXAMPLE

At 3x leverage, a -20% asset move implies about -60% equity before costs; recovery then requires +150% on remaining equity.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Backtests should include financing rate, margin rule, liquidation logic, exposure caps, and rebalancing frequency.

CONFUSIONS TO AVOID

Leverage is not just multiplying final CAGR. It changes the whole sequence and can cause termination before recovery.

RELATED TERMS / QUICK RECALL

Exposure, margin, financing, ruin risk, leveraged ETF decay.

CORE MATH / BEGINNER / FINANCIAL GROWTH

Sequence Risk and Path Dependency

ONE-SENTENCE MEANING

Sequence risk is the risk that the order of returns harms the outcome even when average return looks acceptable.

MEMORY JOGGER

Think: same ingredients, different order, different survival result.

FORMULA / KEY MECHANICS

```
Path-dependent outcome: W_T depends on ordered list (r_1, r_2, ..., r_T), not only on mean(r)
When cash flows, leverage, drawdowns, withdrawals, or risk limits are present, the sequence
order can dominate the average return.
```

REFERENCE TABLE

Path-dependent feature	Why order matters	Diagnostic
Withdrawals	Losses early reduce capital base	Failure probability.
Leverage	Losses can force liquidation	Ruin rate.
Stop rules	Bad early run can shut system down	Stopped-path analysis.

WHY IT MATTERS IN QUANT RESEARCH

Retirement withdrawals, leveraged strategies, stop-outs, and capital allocation rules are all highly sequence-sensitive.

SIMPLE MARKET / TRADING EXAMPLE

Two portfolios may have the same set of annual returns, but the one with large early losses can fail under withdrawals while the other survives.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Stress test by permuting return order, bootstrapping blocks, and simulating poor early regimes. Evaluate survival, not just average terminal wealth.

CONFUSIONS TO AVOID

Do not judge a path-dependent system using only average return and volatility. Order is part of the system.

RELATED TERMS / QUICK RECALL

Path dependency, withdrawal risk, bootstrapping, stress test.

Return Streams as Sequences

ONE-SENTENCE MEANING

A return stream is a time-indexed sequence of strategy or asset returns.

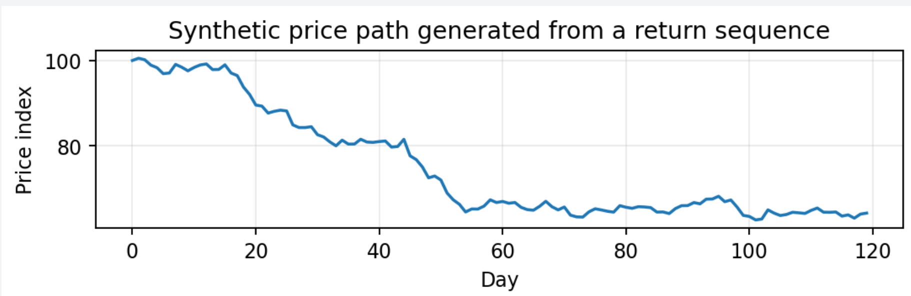
MEMORY JOGGER

Think: the strategy speaks one period return at a time.

FORMULA / KEY MECHANICS

```
Return stream: r_1, r_2, ..., r_T Equity: W_t = W_{t-1}(1+r_t) A return stream can be raw asset return, signal return, portfolio return, factor return, or net strategy return. Its index, frequency, and construction rules define what it means.
```

VISUAL / DIAGRAM



Use the visual as a quick check for the formula and path behavior.

WHY IT MATTERS IN QUANT RESEARCH

Everything downstream depends on the return stream: Sharpe, drawdown, CAGR, turnover, correlation, risk, and portfolio allocation.

SIMPLE MARKET / TRADING EXAMPLE

A daily market-neutral strategy might produce r_t values for each trading day after subtracting borrow and transaction costs.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Version return streams with metadata: universe, weights, rebalance rule, execution assumption, cost model, and survivorship handling.

CONFUSIONS TO AVOID

Do not mix return streams with different frequencies or missing-day conventions without alignment.

RELATED TERMS / QUICK RECALL

Time series, strategy return, factor return, performance analytics.

Rolling Windows

ONE-SENTENCE MEANING

A rolling window computes a statistic over the most recent fixed number of observations.

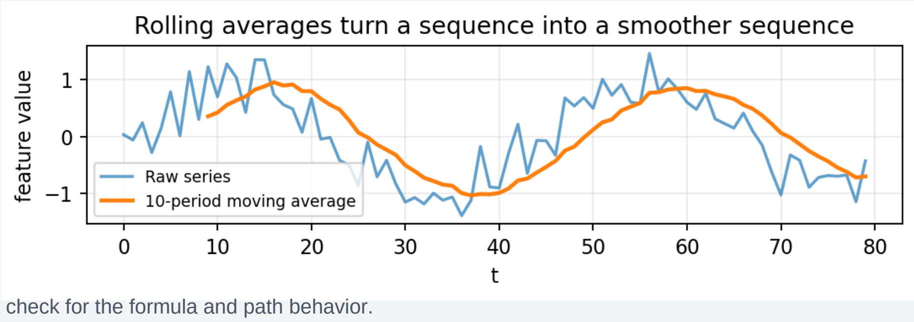
MEMORY JOGGER

Think: a moving local lens over a sequence.

FORMULA / KEY MECHANICS

```
Rolling mean_t = (1/k) x sum_{i=0..k-1} x_{t-i} The window drops old values and adds new ones as time advances. Rolling means, volatility, correlations, beta, and z-scores are standard examples.
```

VISUAL / DIAGRAM



Use the visual as a quick check for the formula and path behavior.

WHY IT MATTERS IN QUANT RESEARCH

Rolling features let models adapt to changing regimes while avoiding use of future data.

SIMPLE MARKET / TRADING EXAMPLE

A 20-day rolling volatility at t should use returns from t-19 through t, then be shifted if trading at the next open.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

In Python, use rolling(window).mean(), rolling(window).std(), and then shift features when needed to avoid same-bar leakage.

CONFUSIONS TO AVOID

A rolling statistic calculated with current close cannot be traded at that same close unless your execution model allows it.

RELATED TERMS / QUICK RECALL

Lookback, moving average, rolling volatility, feature engineering.

CORE MATH / BEGINNER / QUANT RESEARCH SYSTEM

Moving Averages as Sequence Filters

ONE-SENTENCE MEANING

A moving average transforms a noisy sequence into a smoother sequence.

MEMORY JOGGER

Think: blur the short-term noise to see slower movement.

FORMULA / KEY MECHANICS

```
SMA_t = (x_t + x_{t-1} + ... + x_{t-k+1}) / k Simple moving averages weight each value in the window equally. Longer windows smooth more but react more slowly.
```

REFERENCE TABLE

Window length	Benefit	Tradeoff
Short	Fast reaction	Noisy and whipsaw-prone.
Medium	Balanced	Still parameter-sensitive.
Long	Smooth regime view	Lag and slow exits.

WHY IT MATTERS IN QUANT RESEARCH

Moving averages are used for trend features, signal smoothing, volatility estimates, and baseline comparisons.

SIMPLE MARKET / TRADING EXAMPLE

A 50-day moving average crossing above a 200-day moving average is a sequence relationship between two smoothed price paths.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Treat moving averages as features requiring lag discipline, parameter validation, and regime testing.

CONFUSIONS TO AVOID

A moving average is not predictive by itself. It is a transformation; its value depends on signal design and validation.

RELATED TERMS / QUICK RECALL

SMA, smoothing, trend filter, lag.

CORE MATH / BEGINNER / QUANT RESEARCH SYSTEM

Exponentially Weighted Sequences

ONE-SENTENCE MEANING

An exponentially weighted sequence gives recent observations more weight and older observations less weight by geometric decay.

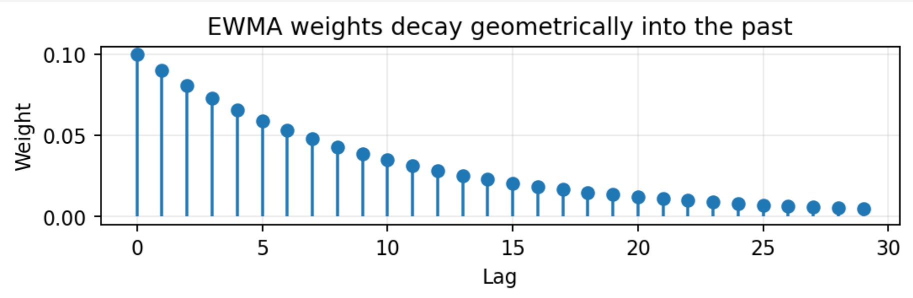
MEMORY JOGGER

Think: memory that fades by a constant percent each step.

FORMULA / KEY MECHANICS

`m_t = alpha x x_t + (1-alpha) x m_{t-1} EWMA uses a recursive blend of the new value and the previous estimate. The decay parameter controls how quickly old information fades.`

VISUAL / DIAGRAM



Use the visual as a quick check for the formula and path behavior.

WHY IT MATTERS IN QUANT RESEARCH

Used for volatility estimation, trend smoothing, signal decay, covariance estimation, and risk models.

SIMPLE MARKET / TRADING EXAMPLE

If alpha is 0.10, the new observation gets 10% weight and the previous estimate carries 90% forward.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Use EWMA when you want adaptive estimates without hard window cutoffs. Store alpha, halflife, or span so the decay is reproducible.

CONFUSIONS TO AVOID

Different libraries parameterize EWMA differently. Alpha, span, center of mass, and halflife can describe the same idea in different forms.

RELATED TERMS / QUICK RECALL

EWMA, decay, halflife, recursive filter.

CORE MATH / BEGINNER / QUANT RESEARCH SYSTEM

Signal Decay and Half-Life

ONE-SENTENCE MEANING

Signal half-life is the time it takes for a signal effect or weight to decay by half.

MEMORY JOGGER

Think: how long the signal stays alive.

FORMULA / KEY MECHANICS

```
If weight after h steps is 0.5, then decay^h = 0.5 h = ln(0.5)/ln(decay) If an effect decays geometrically, half-life translates a decay ratio into a more intuitive time unit.
```

REFERENCE TABLE

Half-life	Research implication	Risk
Short	Trade quickly; costs matter more	Signal may vanish before execution.
Medium	Rebalance periodically	Capacity and turnover tradeoff.
Long	Can hold longer	May be regime-specific.

WHY IT MATTERS IN QUANT RESEARCH

Half-life matters for holding period, rebalance frequency, feature freshness, alpha decay, and execution urgency.

SIMPLE MARKET / TRADING EXAMPLE

A news signal with a 1-day half-life should not be treated like a valuation signal that may persist for months.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Estimate decay by tracking forward returns after signal formation. Compare performance by lag bucket.

CONFUSIONS TO AVOID

A long half-life estimate from a short sample may be noise. Confirm across regimes and assets.

RELATED TERMS / QUICK RECALL

Alpha decay, holding period, rebalance frequency, EWMA.

Backtest Equity Curve

ONE-SENTENCE MEANING

A backtest equity curve is the compounded wealth sequence produced by a strategy return stream.

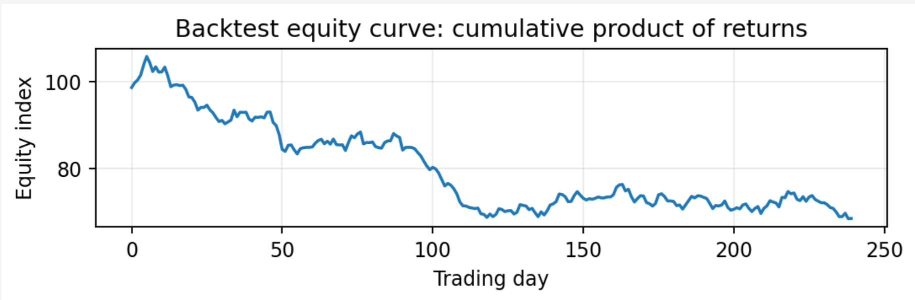
MEMORY JOGGER

Think: the strategy account balance through time.

FORMULA / KEY MECHANICS

`Equity_t = Equity_{t-1} x (1 + net_return_t) The equity curve is built recursively from net returns. It is the source for CAGR, drawdown, underwater periods, and capital-at-risk diagnostics.`

VISUAL / DIAGRAM



Use the visual as a quick check for the formula and path behavior.

WHY IT MATTERS IN QUANT RESEARCH

A strategy is not just a final number; the equity path reveals whether the strategy was smooth, fragile, or exposed to unacceptable losses.

SIMPLE MARKET / TRADING EXAMPLE

Starting at 100, returns [2%, -1%, 3%] produce equity [102.00, 100.98, 104.01].

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Always compute equity after transaction costs and after realistic execution timing. Keep gross and net curves separate.

CONFUSIONS TO AVOID

A beautiful equity curve can still be overfit. Combine curve analysis with out-of-sample validation and robustness checks.

RELATED TERMS / QUICK RECALL

Cumulative return, drawdown, net performance, backtest report.

CORE MATH / BEGINNER / QUANT RESEARCH SYSTEM

Parameter Sweeps as Grid Sequences

ONE-SENTENCE MEANING

A parameter sweep evaluates a strategy across an ordered grid of parameter values.

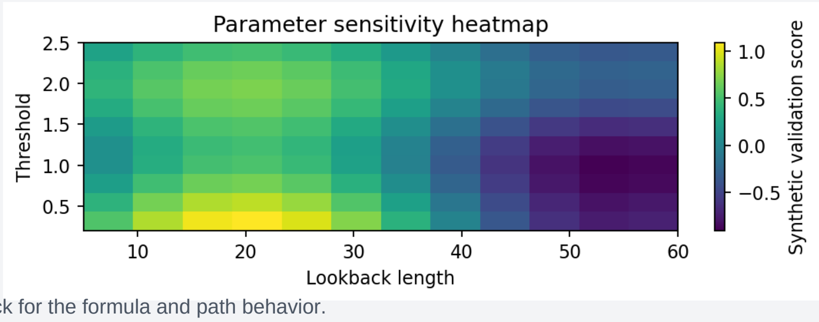
MEMORY JOGGER

Think: a sequence of experiments instead of one favorite setting.

FORMULA / KEY MECHANICS

```
Parameter grid: p in {p_1, p_2, ..., p_k}; evaluate score(p_i) for each setting Parameters
can form arithmetic sequences, geometric sequences, or grids. The pattern of scores across
the grid is often more important than the single best value.
```

VISUAL / DIAGRAM



Use the visual as a quick check for the formula and path behavior.

WHY IT MATTERS IN QUANT RESEARCH

Robust strategies usually show broad stable regions, not one isolated spike surrounded by failure.

SIMPLE MARKET / TRADING EXAMPLE

Testing moving-average windows [5, 10, 20, 40, 80] creates a geometric-ish sweep over lookback scale.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Store every run with parameter values, data slice, costs, objective, and validation period. Plot sensitivity heatmaps.

CONFUSIONS TO AVOID

Choosing the best parameter after looking at all results is optimization bias unless validated properly.

RELATED TERMS / QUICK RECALL

Grid search, sensitivity, robustness, overfitting.

CORE MATH / BEGINNER / QUANT RESEARCH SYSTEM

Monte Carlo Wealth Paths

ONE-SENTENCE MEANING

Monte Carlo simulation creates many possible future sequences and studies the distribution of outcomes.

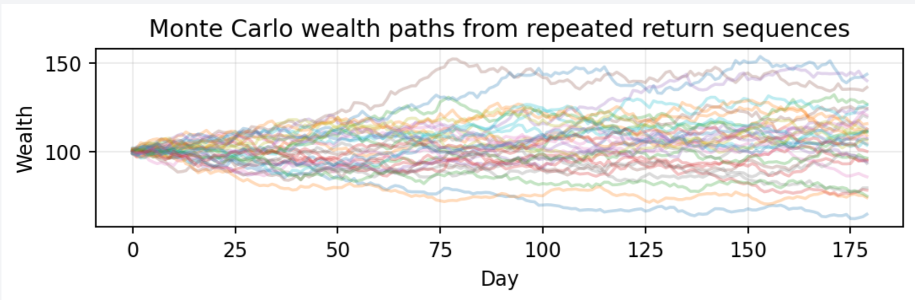
MEMORY JOGGER

Think: many possible paths, not one forecast line.

FORMULA / KEY MECHANICS

```
For each path j: W_{t,j} = W_{t-1,j} x (1 + r_{t,j}) Each simulation path is a sequence of sampled or modeled returns. Compounding each path creates a terminal wealth distribution and path-risk metrics.
```

VISUAL / DIAGRAM



Use the visual as a quick check for the formula and path behavior.

WHY IT MATTERS IN QUANT RESEARCH

Monte Carlo helps estimate ruin probability, drawdown risk, contribution outcomes, and uncertainty around long-term projections.

SIMPLE MARKET / TRADING EXAMPLE

Sampling daily returns with replacement can generate thousands of alternate equity paths from historical behavior.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Use simulation assumptions explicitly: sampling method, distribution, autocorrelation, regime model, and rebalancing rule.

CONFUSIONS TO AVOID

Monte Carlo output is only as good as the assumptions. Random-looking paths can still miss structural breaks and liquidity crises.

RELATED TERMS / QUICK RECALL

Simulation, bootstrap, scenario analysis, terminal wealth.

CORE MATH / BEGINNER / QUANT RESEARCH SYSTEM

Regime Stability Across Sequences

ONE-SENTENCE MEANING

Regime stability checks whether a concept or strategy behaves similarly across different market environments.

MEMORY JOGGER

Think: does the rule survive different weather?

FORMULA / KEY MECHANICS

```
Stability check: compare metric(sequence slice A), metric(slice B), metric(slice C) A
sequence can be split into trend, chop, high-volatility, low-volatility, crisis, and calm
periods. Metrics are then compared across slices.
```

REFERENCE TABLE

Regime slice	What to inspect	Failure sign
High volatility	Drawdown and turnover	Costs explode.
Low volatility	Signal strength	Returns vanish.
Trend	Directional exposure	Beta masquerades as alpha.
Mean reversion	Whipsaw risk	Signal flips too late.

WHY IT MATTERS IN QUANT RESEARCH

A strategy that only works in one regime may still be useful, but it needs regime detection or allocation control.

SIMPLE MARKET / TRADING EXAMPLE

A momentum strategy may work in trending markets and bleed during sideways mean-reverting periods.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Report performance by period, asset group, volatility bucket, and macro regime. Look for consistent signs, not identical numbers.

CONFUSIONS TO AVOID

Do not average regimes together and assume the combined result is stable. One regime can carry the whole backtest.

RELATED TERMS / QUICK RECALL

Regime, robustness, subsample, walk-forward validation.

Implementation Checklist

ONE-SENTENCE MEANING

A sequence-aware implementation makes ordering, timing, and compounding explicit in code.

MEMORY JOGGER

Think: every calculation needs a timestamp and an availability rule.

FORMULA / KEY MECHANICS

```
Minimal pipeline: raw data -> aligned sequence -> returns -> signal -> position -> net returns -> equity -> diagnostics The implementation goal is not just to compute formulas; it is to prevent ambiguous data flow and invalid performance measurement.
```

REFERENCE TABLE

Check	Why it matters	Pass condition
Index sorted	Sequence order valid	Timestamps increase.
No future data	Avoid look-ahead	Features lagged correctly.
Compounding correct	Equity valid	cumprod equals loop.
Costs applied	Net results realistic	Gross >= net usually.

WHY IT MATTERS IN QUANT RESEARCH

Most beginner quant errors are sequence errors: wrong shift, wrong compounding, wrong resampling, wrong cash-flow timing, or wrong cost timing.

SIMPLE MARKET / TRADING EXAMPLE

Before trusting a backtest, inspect the first 20 rows with prices, returns, signal, position, trade, cost, and equity.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Build tests for formula identities: log returns telescope, equity cumprod matches iterative update, drawdown stays <= 0.

CONFUSIONS TO AVOID

Do not hide sequence logic inside black-box library calls until you understand the index alignment.

RELATED TERMS / QUICK RECALL

Testing, validation, data engineering, backtest audit.

Common Failure Modes

ONE-SENTENCE MEANING

Failure modes are recurring mistakes that make sequence, series, and growth calculations misleading.

MEMORY JOGGER

Think: these are the bugs that make a fake edge look real.

FORMULA / KEY MECHANICS

```
Audit rule: every value must answer "known when? used when? applied how?" The most dangerous
errors are silent: the code runs, the chart looks plausible, and the conclusion is wrong.
```

REFERENCE TABLE

Mistake	Symptom	Fix
Adding simple returns	Cumulative return too high or odd	Use product(1+r)-1.
Unshifted feature	Suspiciously strong performance	Lag features before trading.
Wrong cash-flow timing	Deposits boost returns	Separate performance from contributions.
Ignoring costs	High-turnover alpha disappears live	Apply costs per trade.

WHY IT MATTERS IN QUANT RESEARCH

Failure-mode thinking is essential in quant because a small timing or compounding bug can dominate measured alpha.

SIMPLE MARKET / TRADING EXAMPLE

A strategy using today close to generate a signal and also execute at today close may accidentally trade with future information.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Add automated checks and manual audit rows to every research notebook or pipeline run.

CONFUSIONS TO AVOID

Do not only debug syntax. Debug assumptions: what was known, when it was known, and how wealth changed.

RELATED TERMS / QUICK RECALL

Look-ahead bias, compounding error, resampling bug, overfitting.

Formula Summary Sheet

ONE-SENTENCE MEANING

The formula summary collects the core equations needed for fast recall.

MEMORY JOGGER

Think: the minimum viable math sheet for this topic.

FORMULA / KEY MECHANICS

```
a_n = a_1 + (n-1)d | a_n = a_1 q^(n-1) | S_n = sum a_i | W_T = W_0 product(1+r_t) | CAGR = (W_T/W_0)^(1/years)-1 | PV = FV/(1+r)^n Use this page as a memory anchor before returning to individual concept cards for explanations and examples.
```

REFERENCE TABLE

Formula	Use	Main warning
product(1+r_t)-1	Cumulative simple return	Do not add returns.
sum log(1+r_t)	Cumulative log return	Convert back with exp.
Drawdown = W/peak - 1	Path risk	Needs running peak.
Annuity FV	Recurring equal payments	Assumes constant rate.

WHY IT MATTERS IN QUANT RESEARCH

These equations cover sequences, series, compounding, discounting, returns, CAGR, drawdown, and recurring payments.

SIMPLE MARKET / TRADING EXAMPLE

If you can explain each equation in plain English and implement it on a small dataset, you understand the operational core.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Turn this page into unit tests in your research code: give known inputs and assert known outputs.

CONFUSIONS TO AVOID

Memorizing formulas without knowing when they apply is fragile. Pair each equation with its assumptions.

RELATED TERMS / QUICK RECALL

Cheat sheet, formula block, implementation tests, review.

Quick Review Questions

ONE-SENTENCE MEANING

Review questions force active recall of the difference between ordered values, summed values, and compounded values.

MEMORY JOGGER

Think: do not just reread - answer.

FORMULA / KEY MECHANICS

```
Decision rule: cash flows often add; wealth returns usually multiply; log growth sums after
log transform. The goal is to identify whether a problem needs a sequence, a sum, a product,
a discount factor, or a recursive update.
```

REFERENCE TABLE

Question	Expected move	Answer sketch
Sequence or series?	Check whether order or total is asked	Price path = sequence; cumulative costs = series.
Add or multiply returns?	Use wealth example	Multiply growth factors.
CAGR or average return?	Ask if realized multi-period growth is needed	Use CAGR/geometric.
PV or FV?	Direction in time	PV discounts backward; FV compounds forward.

WHY IT MATTERS IN QUANT RESEARCH

These questions are designed to expose the common beginner confusion points in financial growth math.

SIMPLE MARKET / TRADING EXAMPLE

If returns are +20% and -20%, what is the final wealth from $100? If log returns sum to 0.10, what is the growth multiplier?

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Use the questions as prompts for small Python functions and unit tests. Each answer should be reproducible numerically.

CONFUSIONS TO AVOID

If you cannot decide between addition and multiplication, translate the problem into dollars of wealth and step through time.

RELATED TERMS / QUICK RECALL

Active recall, unit testing, formula selection, diagnostics.

Index of Related Terms

ONE-SENTENCE MEANING

This index ties the manual vocabulary into one connected map for later lookup.

MEMORY JOGGER

Think: a local search map for the PDF.

FORMULA / KEY MECHANICS

```
Core map: ordered data -> transformed sequence -> aggregate series/product -> growth metric
-> risk diagnostic Related terms matter because the same math appears under different names
in trading, portfolio analytics, valuation, and system design.
```

REFERENCE TABLE

Lookup term	Go to concept	Connection
cumprod	Cumulative Return as Product	Turns returns into wealth.
cumsum	Partial Sums	Turns additive terms into running totals.
halflife	Signal Decay and Half-Life	Converts decay to time intuition.
fee drag	Fees, Taxes, and Slippage	Shows friction compounding.
path risk	Sequence Risk	Order affects survival.

WHY IT MATTERS IN QUANT RESEARCH

When confused, search for the operational object: sequence, sum, product, recursive state, cash flow, discount factor, or diagnostic metric.

SIMPLE MARKET / TRADING EXAMPLE

A drawdown page is also about running maxima, path dependency, compounding, and equity curves.

WHERE IT FITS IN THE RESEARCH SYSTEM / PYTHON IDEA

Use this index to decide which concept page to revisit when building a feature, performance report, or backtest module.

CONFUSIONS TO AVOID

Do not isolate formulas. Most quant bugs happen at the boundary between concepts, such as returns into equity or signals into positions.

RELATED TERMS / QUICK RECALL

Lookup, concept map, review, research system vocabulary.